# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## ANSWER KEY & MARKING SCHEME SET A

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 3 : 4 | Divide both by HCF 6 |
| 2 | **(b)** 3 : 4 | Divide both by HCF 18 |
| 3 | **(a)** 35 | Product of extremes = product of means: $2x = 70$ |
| 4 | **(d)** ₹ 400 | A gets 5/9 of 720 |
| 5 | **(b)** 2500 m | Unit conversion |
| 6 | **(b)** 0.15 m | Unit conversion |
| 7 | **(c)** 12 | Multiply both by 3 |
| 8 | **(b)** 420 km | 60 km/h |
| 9 | **(b)** 2 : 7 | 12 : 42 |
| 10 | **(c)** 17.5 km | 3.5 × 5 |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Comparison of two quantities by division (1); $2 : 3$ (1). | 1+1 |
| 12 | $75 : 300 = 1 : 4$ (2). | 2 |
| 13 | Yes, $4 \times 27 = 108 = 9 \times 12$ (2). | 2 |
| 14 | Shares ₹ 450 and ₹ 750 (2). | 2 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Total parts $= 5$ (1); one part $=$ ₹$300$ (1); shares ₹ 600, ₹ 900 (1). | 1+1+1 |
| 16 | Total parts $= 8$ (1); one part $=$ ₹$300$ (1); shares ₹ 900, ₹ 1500 (1). | 1+1+1 |
| 17 | Total parts $= 7$ (1); one part $=$ ₹$600$ (1); shares ₹ 2400, ₹ 1800 (1). | 1+1+1 |
| | **OR:** Total parts $= 6$ (1); one part $=$ ₹$300$ (1); shares ₹ 300, ₹ 600, ₹ 900 (1). | 1+1+1 |
| 18 | Total parts $= 11$ (1); one part $=$ ₹$300$ (1); shares ₹ 600, ₹ 900, ₹ 1800 (1). | 1+1+1 |
| | **OR:** Total parts $= 14$ (1); one part $=$ ₹$200$ (1); shares ₹ 600, ₹ 800, ₹ 1400 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) Ratio is a comparison by division; proportion is equality of two ratios (2). (ii) $x = 9$ (1.5). (iii) $6 \times 15 = 90 = 9 \times 10$, yes (1.5). | 2+1½+1½ |
| | **OR:** (i) ₹ 1500, ₹ 2250, ₹ 3750 (3). (ii) A 2000 : B 2250 : C 3250 = 8 : 9 : 13 (2). | 3+2 |
| 20 | (i) 3 : 2 : 1 (1). (ii) 750 g, 500 g, 250 g (2). (iii) 20 people (2). | 1+2+2 |
| | **OR:** (i) 185 km (2). (ii) $1 : 2500000$ (3). | 2+3 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

## ANSWER KEY & MARKING SCHEME SET B

Chapter 7: **Proportional Reasoning-1** | Class 8 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 3 : 4 | Divide both by HCF 15 |
| 2 | **(b)** 1 : 4 | Divide both by HCF 25 |
| 3 | **(c)** 18 | Product of extremes = product of means: $5x = 90$ |
| 4 | **(a)** ₹ 630 | A gets 7/10 of 900 |
| 5 | **(b)** 180 minutes | Unit conversion |
| 6 | **(d)** 1.5 hours | Unit conversion |
| 7 | **(a)** have the same value when simplified | Proportion |
| 8 | **(c)** 12 L | 3/5 of 20 |
| 9 | **(a)** 45 | 9 units |
| 10 | **(d)** 15 | Multiply by 3 |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | 3600 m (1); 0.45 L (1). | 1+1 |
| 12 | Inverse relation: $8 \times 15/12 = 10$ days (2). | 2 |
| 13 | $x = 25$ (2). | 2 |
| 14 | 25 dogs (2). | 2 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Let numbers be $3x$, $2x$; $5x = 40$ (1); $x = 8$ (1); numbers 24 and 16 (1). | 1+1+1 |
| 16 | Let numbers be $5x$, $3x$; $8x = 64$ (1); $x = 8$ (1); numbers 40 and 24 (1). | 1+1+1 |
| 17 | Let numbers be $4x$, $5x$; $9x = 72$ (1); $x = 8$ (1); numbers 32 and 40 (1). | 1+1+1 |
| | **OR:** Let numbers be $7x$, $3x$; $10x = 50$ (1); $x = 5$ (1); numbers 35 and 15 (1). | 1+1+1 |
| 18 | Let numbers be $2x$, $7x$; $9x = 90$ (1); $x = 10$ (1); numbers 20 and 70 (1). | 1+1+1 |
| | **OR:** Let numbers be $3x$, $8x$; $11x = 66$ (1); $x = 6$ (1); numbers 18 and 48 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) 16 days (2.5). (ii) 30 men (2.5). | 2½+2½ |
| | **OR:** (i) 300 and 180 (2). (ii) 280 : 200 = 7 : 5 (2). (iii) Ratio changes as numbers shift (1). | 2+2+1 |
| 20 | (i) 9 : 12 : 14 (2). (ii) Total parts 35, so 45, 60 and 70 (3). | 2+3 |
| | **OR:** (i) $36 \times 1000/3600 = 10$ m/s (2). (ii) 18 km/h (3). | 2+3 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.